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ANNAM.AI · COMMUNITY PREDICTION COMPETITION · PRIVATE · A YEAR AGO

Late Submission

Soil Classification

An initial challenge to classify soil types

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Overview

Soil Image Classification Challenge

The Soil Image Classification Challenge is a machine learning competition organised by Annam.ai at IIT Ropar, serving as an initial task for shortlisted hackathon participants. Competitors will build models to classify each soil image into one of four categories: Alluvial soil, Black soil, Clay soil, or Red soil. Final submissions are due by May 25, 2025, 11:59 PM IST. Be sure to submit well before the deadline to avoid server overload or last-minute issues.

- Task: Classify each provided soil image into one of the four soil types (Alluvial, Black, Clay, Red).
- Deadline: May 25, 2025, 11:59 PM IST (please plan ahead to avoid system delays).
- Note: Do not wait until the last minute to submit; late traffic may lead to upload failures.

Start May 19, 2025

Close May 25, 2025

Description

Competition Host
Annam.ai

Prizes & Awards
Kudos
Does not award Points or Medals

Participation
502 Entrants
389 Participants
169 Teams
514 Submissions

Tags
Custom Metric

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Soil Classification

Late Submission

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Description

Soil plays a foundational role in agriculture and the environment. Healthy soils support plant growth and food production, so being able to quickly and accurately identify soil type is valuable for crop management, land use planning, and environmental assessment. In fact, researchers note that soil classification is "a burgeoning field" with importance across farming, geology, and engineering. In this challenge, participants will apply computer vision and machine learning techniques to automate soil classification from images. Participants will train a model (typically a convolutional neural network) on labeled soil images so that it can predict the soil type of new images. Each image has a ground-truth label (one of the four soil types). The model should learn from features like color, texture, and patterns in the soil photographs. During development, you should split data into training/validation sets and tune your model accordingly. Input images in the dataset may vary in size and resolution. In preprocessing, you will standardise the image dimensions (for example, by resizing or cropping images to 224×224 pixels) so they can be processed by common neural network architectures. You may also normalise pixel values and apply augmentation as needed. Ensuring consistent formatting and sufficient data cleaning (e.g. handling brightness or background variations) will help the model learn effectively. Finally, clean and well-documented code is crucial.

- Organise your solution in a clear notebook or script: use logical variable names, break the workflow into understandable steps, and include comments explaining your approach.
- Clear explanations and reproducible code will not only help the reviewers follow your work but also contribute to better overall performance.

Evaluation

The competition metric is balanced classification performance, as defined below.

- Your submitted model will be scored by computing the F1-score for each of the four soil categories, and then taking the minimum of those four F1-scores. In other words, the final score is the lowest per-class F1:

Metric: Minimum F1-score across all classes (maximise this). This encourages models to perform well on every soil type, not just on average.

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Soil Classification

Evaluation

The competition metric is balanced classification performance, as defined below.

- Your submitted model will be scored by computing the F1-score for each of the four soil categories, and then taking the minimum of those four F1-scores. In other words, the final score is the lowest per-class F1:

Metric: Minimum F1-score across all classes (maximise this). This encourages models to perform well on every soil type, not just on average.

Why F1: The F1-score is the harmonic mean of precision and recall, giving a single number that balances both. It ranges from 0 (worst) to 1 (best). By using the minimum class F1, the competition encourages balanced accuracy across all soil types.

Maximising the minimum F1 thus means improving the worst-performing class as much as possible.

Additional assessment: Besides the numeric score, the final evaluation will include a review of submitted notebooks. Solutions should have clear logic and thorough comments so that reviewers (and future users) can easily follow your workflow. Code quality, organisation, documentation and other deliverables as were discussed during orientation are part of the judging criteria.

The F1-score penalises classifiers that have an imbalance between precision and recall. By focusing on the minimum class F1-score, this competition rewards models that perform robustly across all soil categories rather than favouring any single category. Interpretation: An F1 score of 1.0 for a class means the model achieved perfect precision and recall on that class. An F1 score of 0 indicates complete failure (either precision or recall is zero) for that class. The final score is the worst of the four F1 scores, so a strong entry will achieve high F1 on every soil type.

- Submission Requirements: Participants must submit their predictions via the competition platform by the deadline. In addition to the prediction CSV, a well-documented code notebook or script must be submitted. The notebook should include the code to preprocess the data, train and evaluate the model, and generate the final predictions. Clear comments and explanations will help ensure your approach is understood and rewarded.

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Soil Classification

An initial challenge to classify soil types

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Dataset Description

This competition provides a dataset of soil images for a multi-class classification task. The goal is to classify each image into one of four soil types: Alluvial soil, Black Soil, Clay soil, or Red soil. The dataset is split into training and test sets to facilitate model development and evaluation. Structure

Training Set:

- Contains images and their corresponding labels.
- Labels are provided in a CSV file with columns: image_id (unique identifier for each image) and soil_type (the soil type label).
- Use this set to train your model.

Test Set:

- Contains images for which you must predict the soil type.
- A CSV file lists the image_id for each test image (no labels provided).
- Submit your predictions for these images in the required format.

Images:

- Organised into separate train/ and test/ directories.
- Image sizes vary, ranging from small to large resolutions. Participants should preprocess images as needed (e.g., resize to a standard size like 224×224 for model compatibility).

Files
1566 files

Size
153.18 MB

Type
jpg, jpeg, png + 3 others

License
MIT

Soil Classification

Classes

The dataset includes four soil types:

- Alluvial soil
- Black Soil
- Clay soil
- Red soil

Notes:

- Ensure your model handles all classes well, as the evaluation metric focuses on balanced performance across classes.
- Preprocess images to account for varying sizes and ensure compatibility with your model.
- Use only the training set for model training. The test set is for evaluation purposes only, and using test images for training violates competition rules.

This dataset provides a foundational challenge for soil classification, with future training levels introducing additional complexities.

soil_classification-2025 (2 directories, 3 files)

test	341 files
train	1222 files
sample_submission.csv	185 B
test_ids.csv	5.84 kB
train_labels.csv	35.19 kB

Data Explorer

153.18 MB

- soil_classification-2025
  - test
  - train
  - sample_submission.csv
  - test_ids.csv
  - train_labels.csv

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Late Submission

Soil Classification

An initial challenge to classify soil types

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Leaderboard

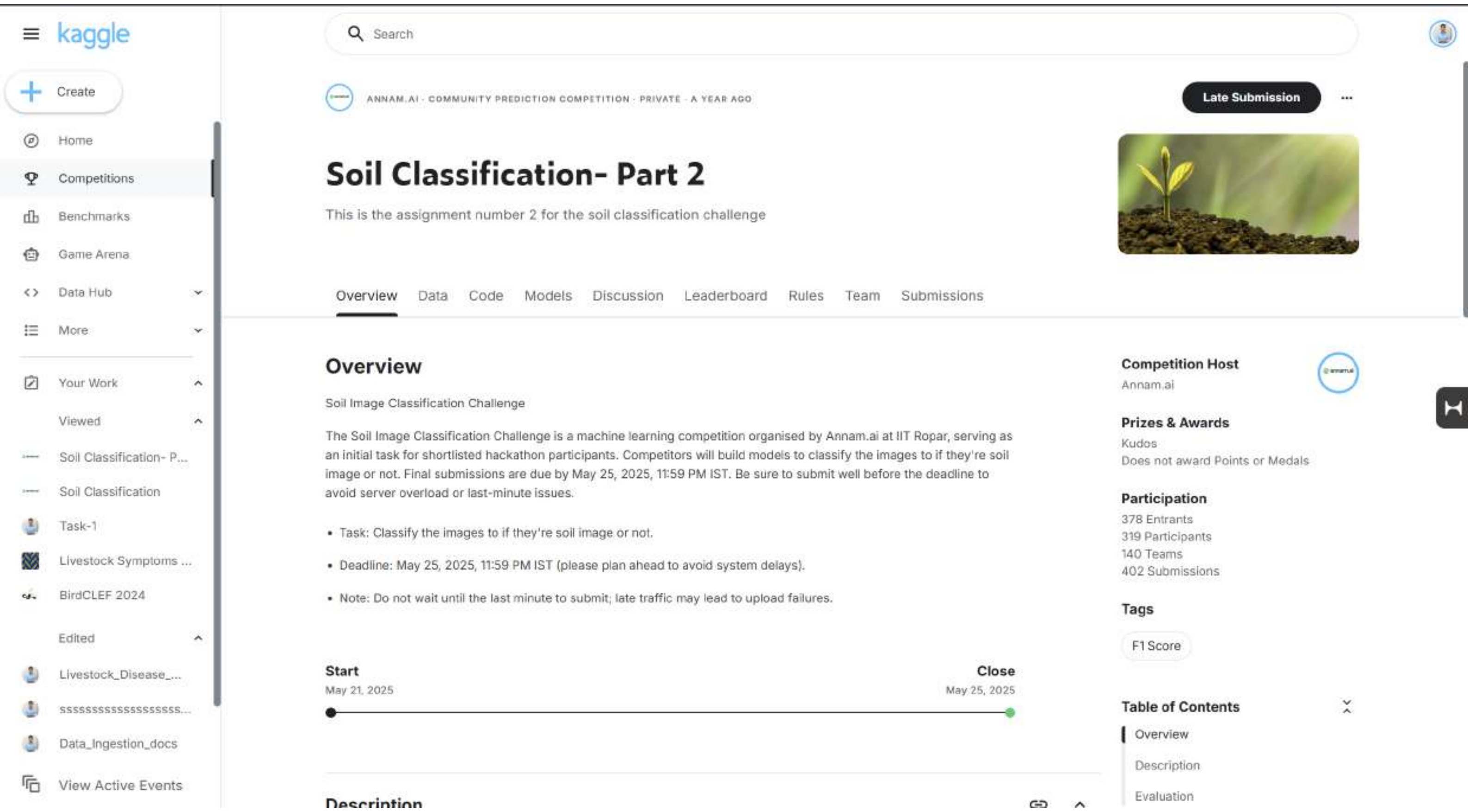
Raw Data Refresh

Virendra

Public Private

The private leaderboard is calculated with approximately 70% of the test data.
This competition has completed. This leaderboard reflects the final standings.

#	△	Team	Members	Score	Entries	Last	Solution
109	▾ 57	Virendra Badgotya		0.9610	4	1y	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Late Submission
Soil Classification- Part 2
This is the assignment number 2 for the soil classification challenge
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Overview
Soil Image Classification Challenge
The Soil Image Classification Challenge is a machine learning competition organised by Annam.ai at IIT Ropar, serving as an initial task for shortlisted hackathon participants. Competitors will build models to classify the images to if they're soil image or not. Final submissions are due by May 25, 2025, 11:59 PM IST. Be sure to submit well before the deadline to avoid server overload or last-minute issues.
• Task: Classify the images to if they're soil image or not.
• Deadline: May 25, 2025, 11:59 PM IST (please plan ahead to avoid system delays).
• Note: Do not wait until the last minute to submit; late traffic may lead to upload failures.
Start
May 21, 2025
Close
May 25, 2025
Description
Competition Host
Annam.ai
Prizes & Awards
Kudos
Does not award Points or Medals
Participation
378 Entrants
319 Participants
140 Teams
402 Submissions
Tags
F1 Score
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Soil Classification- Part 2

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Description

Soil plays a foundational role in agriculture and the environment. Healthy soils support plant growth and food production, so being able to quickly and accurately identify soil type is valuable for crop management, land use planning, and environmental assessment. In fact, researchers note that soil classification is "a burgeoning field" with importance across farming, geology, and engineering. In this challenge, participants will apply computer vision and machine learning techniques to automate soil classification from images. Participants will train a model (typically a convolutional neural network) on labeled soil images so that it can predict the if the new image is a type of soil or not. Each image has a ground-truth label . The model should learn from features like color, texture, and patterns in the soil photographs. During development, you should split data into training/validation sets and tune your model accordingly. Input images in the dataset may vary in size and resolution. In preprocessing, you will standardise the image dimensions (for example, by resizing or cropping images to 224×224 pixels) so they can be processed by common neural network architectures. You may also normalise pixel values and apply augmentation as needed. Ensuring consistent formatting and sufficient data cleaning (e.g. handling brightness or background variations) will help the model learn effectively. Finally, clean and well-documented code is crucial.

- Organise your solution in a clear notebook or script: use logical variable names, break the workflow into understandable steps, and include comments explaining your approach.
- Clear explanations and reproducible code will not only help the reviewers follow your work but also contribute to better overall performance.

Evaluation

The competition metric is classification performance, as defined below.

- Your submitted model will be scored by computing the F1-score for the binary classification

Why F1: The F1-score is the harmonic mean of precision and recall, giving a single number that balances both. It ranges from 0 (worst) to 1 (best). By using the minimum class F1, the competition encourages balanced accuracy across all soil

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Soil Classification- Part 2

better overall performance.

Evaluation

The competition metric is classification performance, as defined below.

- Your submitted model will be scored by computing the F1-score for the binary classification

Why F1: The F1-score is the harmonic mean of precision and recall, giving a single number that balances both. It ranges from 0 (worst) to 1 (best). By using the minimum class F1, the competition encourages balanced accuracy across all soil types.

Additional assessment: Besides the numeric score, the final evaluation will include a review of submitted notebooks. Solutions should have clear logic and thorough comments so that reviewers (and future users) can easily follow your workflow. Code quality, organisation, documentation and other deliverables as were discussed during orientation are part of the judging criteria.

The F1-score penalises classifiers that have an imbalance between precision and recall. By focusing on F1-score, this competition rewards models that perform robustly across all soil categories rather than favouring any single category. Interpretation: An F1 score of 1.0 for a class means the model achieved perfect precision and recall on that class. An F1 score of 0 indicates complete failure (either precision or recall is zero) for that class.

- Submission Requirements: Participants must submit their predictions via the competition platform by the deadline. In addition to the prediction CSV, a well-documented code notebook or script must be submitted. The notebook should include the code to preprocess the data, train and evaluate the model, and generate the final predictions. Clear comments and explanations will help ensure your approach is understood and rewarded.

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Soil Classification- Part 2

This is the assignment number 2 for the soil classification challenge

Overview Data Code Models Discussion Leaderboard Rules Team Submissions

Dataset Description

This competition provides a dataset of soil images for a binary classification. The goal is to classify the images and check if it's a soil image or not. The dataset is split into training and test sets to facilitate model development and evaluation.

Structure

Training Set:

- Contains images and their corresponding labels.
- Labels are provided in a CSV file with columns: image_id (unique identifier for each image) and soil_type (the soil type label).
- Use this set to train your model.

Test Set:

- The images from the test set are not public.
- A CSV file lists the image_id for each test image.
- Submit your predictions for these images in the required format.

Images:

- Organised into separate train/ directory.
- Participants should preprocess images as needed.

Files
2192 files

Size
145.86 MB

Type
jpg, jpeg, png + 3 others

License
Unset

Soil Classification- Part 2

- Organised into separate train/ directory.
- Participants should preprocess images as needed.

Classes

The dataset includes four soil types:

- Alluvial soil
- Black Soil
- Clay soil
- Red soil

Notes:

- Preprocess images to account for varying sizes and ensure compatibility with your model.
- Use only the training set for model training. The test set is for evaluation purposes only

This dataset provides a foundational challenge for soil classification, with future training levels introducing additional complexities.

soil_competition-2025 (2 directories, 3 files)

test	967 files
train	1222 files
sample_submission.csv	172 B
test_ids.csv	35.79 kB
train_labels.csv	23.33 kB

Data Explorer

145.86 MB

- soil_competition-2025
  - test
  - train
  - sample_submission.csv
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Late Submission

Soil Classification- Part 2

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Leaderboard

Raw Data Refresh

Virendr

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#	△	Team	Members	Score	Entries	Last	Solution
50	▾ 12	Virendra Badgotya		0.8666	4	1y	